Terra – Extractors Tool Document

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Extractors Page

A library ready to deploy Pharma extractors for syndicated data sources.

Extractors play a pivotal role in Terra, facilitating the extraction of data from diverse sources. Within the Extractors module, two primary components are present: Admin and Extractors.

Extractor Landing Page

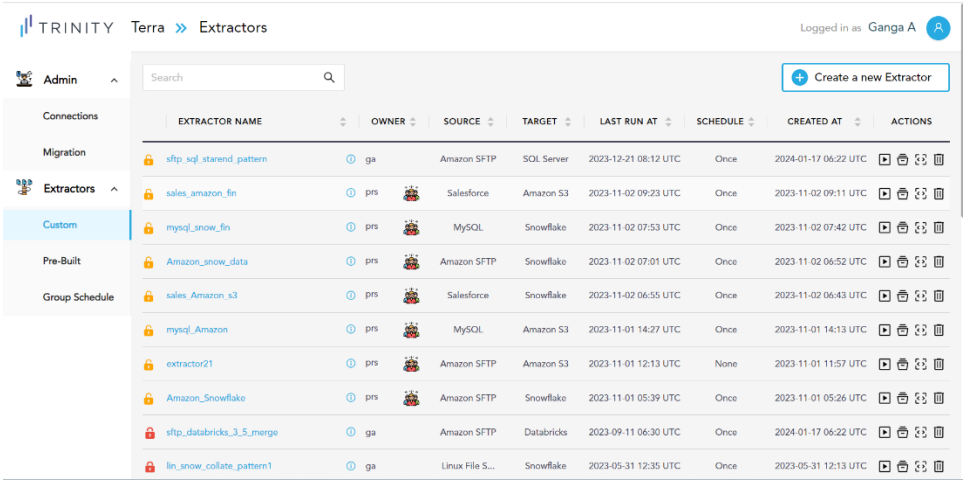
In the landing page, we can see all the extractors that has been created and shared along with four Actions.

1.Run Now

2.Logs

3.Notebook

4.Delete Extractor



1.Run Now: The "Run Now" feature for a workflow provides a way to manually initiate the execution of a workflow outside of its scheduled or triggered run times.

The primary use of "Run Now" is to execute an extractor immediately when needed, without waiting for its regular schedule or triggering event.

In emergency situations or when immediate action is necessary, using "Run Now" allows for a rapid response without waiting for the next scheduled run.

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If a user requires a specific workflow execution outside of the regular schedule, they can trigger it using the "Run Now" option.

2.Logs: Logs show the status of the step-by-step execution of the extractor with the help of airflow dag.

3.Notebook: Notebook code is generated when the user creates the Extractor. Step by step creation of an extractor is depicted in the code.

4.Delete: Deleting an extractor permanently removes it from the tool. Users can delete only the extractors they created, and they cannot delete restricted or public extractors.

Custom

Custom extractors source data from different data sources, such as a custom database, and present it in a form that can be loaded into the desired target data source.

Steps for Creating an Extractor

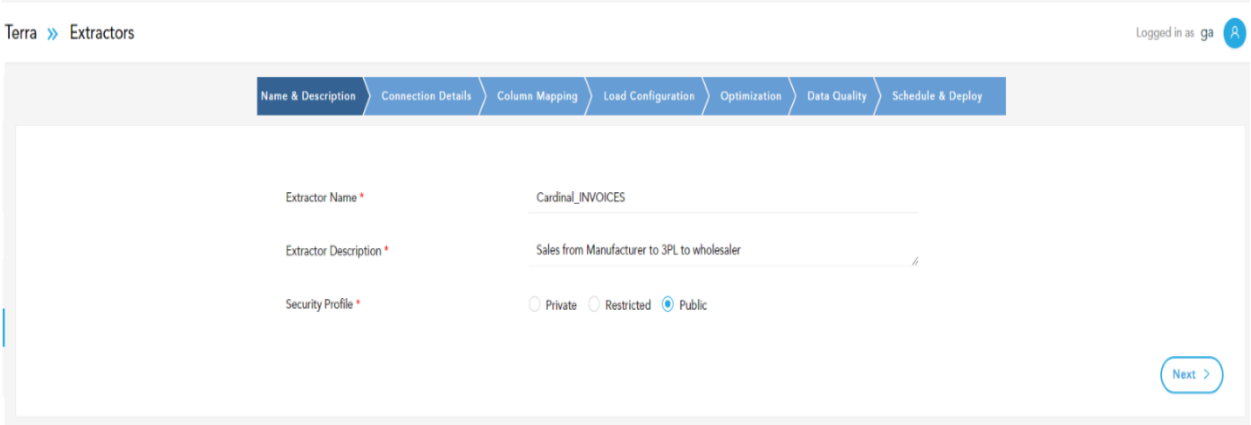
Select Extractors ➔ Custom ➔ Create a New Extractor

1. Enter the name and description of the extractor and choose the security Profile. It will prompt you to enter the extractor name and description. Choose a security profile: Private, Restricted, or Public.

   Private: The extractor is visible only under the user's credentials.

   Restricted: Users can share access with others, allowing changes, scheduling, and deployment by the restricted user.

   Public: The extractor is visible to everyone with access to the Terra application.



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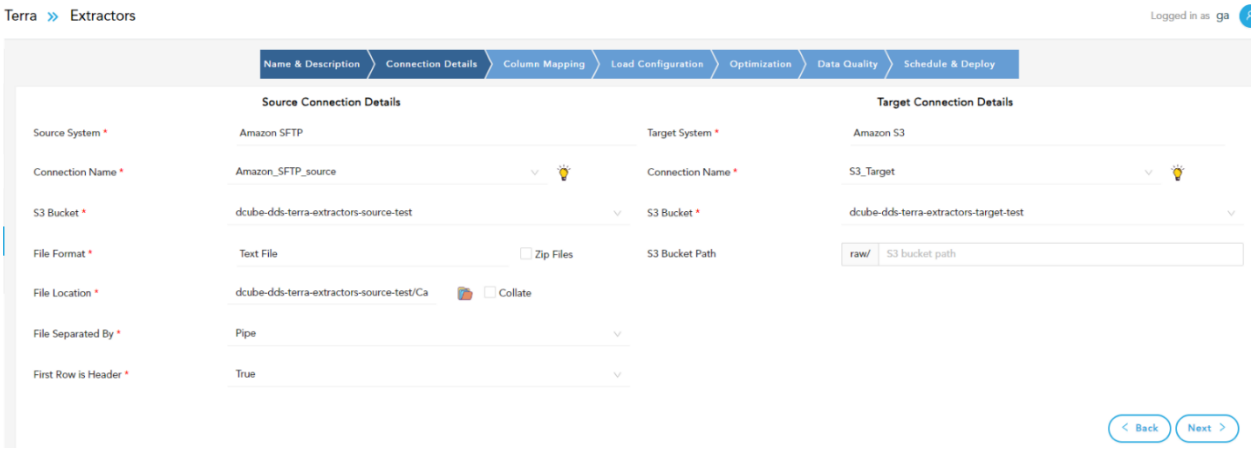
2. Enter the Connection details

a) Enter the source connection details
The bulb indicator validates the connection—glowing for a valid connection and staying unlit for an invalid one. For example, selecting the source system as Amazon SFTP and the connection as Amazon SFTP, the user then chooses the bucket for data loading. Choose the Source file from the location.

b) Choose the header as true or false (Whether first row of data is the header or not)
File format options include CSV, TXT, JSON, XLSX, and XLS. In this case, CSV is selected, and the file path from the chosen S3 bucket is specified. The user can selects the file separator based on the chosen file format (e.g., comma, pipe, or tab). The "First Row Header" option is chosen based on whether the selected file has a header in the first row

c) Enter the target connection details. The user selects the target source system (e.g., Amazon S3), specifies the target connection name, and provides the database name for storing the target data.

d) Enter the target file name and click on Next

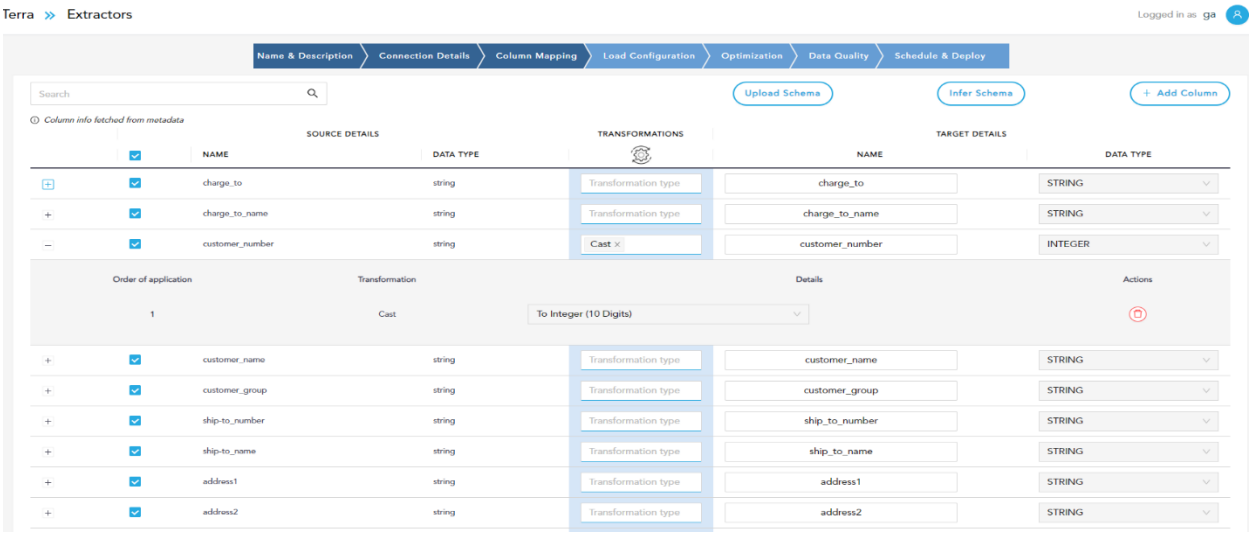


2. Apply transformations on the source data to get the desired target data

In this page user can see the data of the source file which was selected in connection details page. Now the user can change the datatype of the target column and change the target column name, Apply transformations according to data and datatype. User can add new column as well using Add Column.

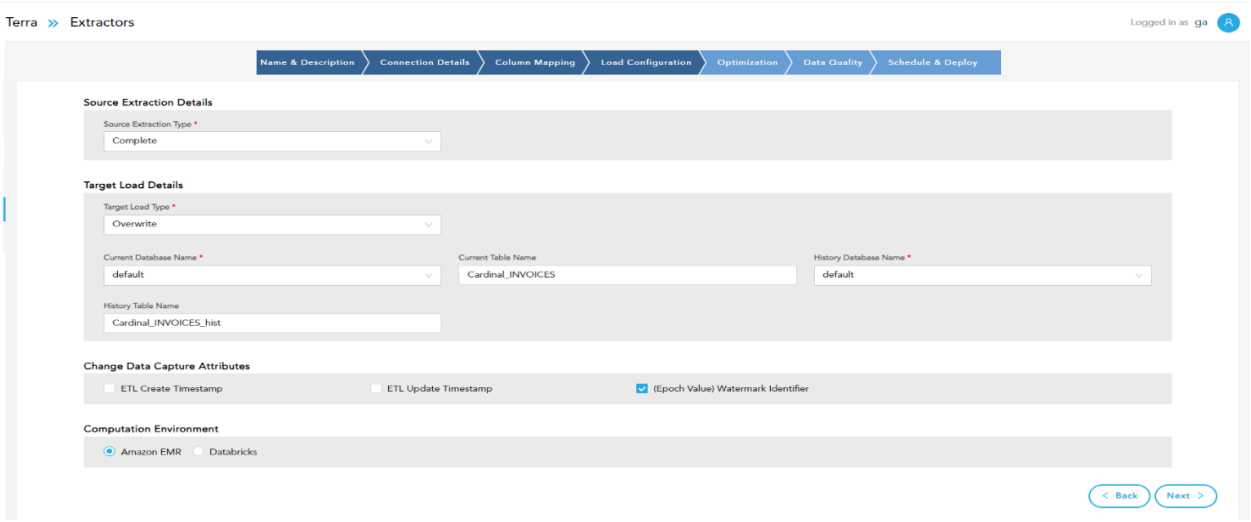
a) If the header is false or if the header is stored in a file, use upload schema button to upload the schema.

b) Infer schema option will be visible for already created extractors which shows any difference in the schema is found and options are there to retain or choose the new schema.

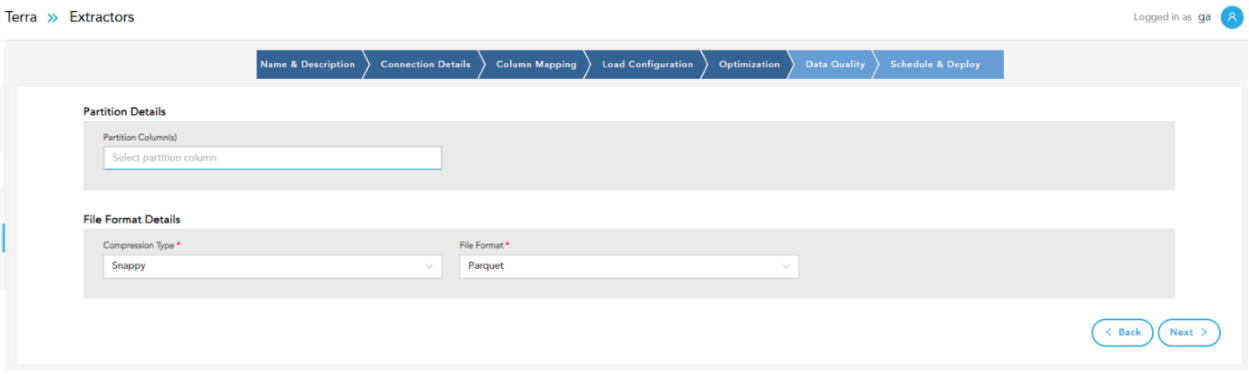


3. Enter the details for Load configuration

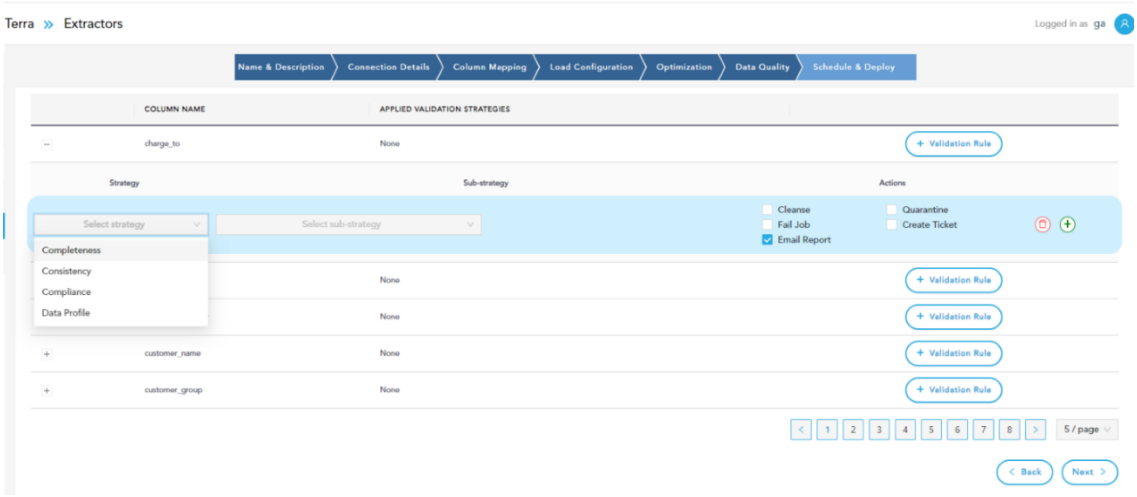
   a. Enter the source extraction type(Complete or Increment).
   b. Enter Target load type (Merge, Append and Overwrite) and other load details.
   c. Choose the computation environment (EMR or Databricks)

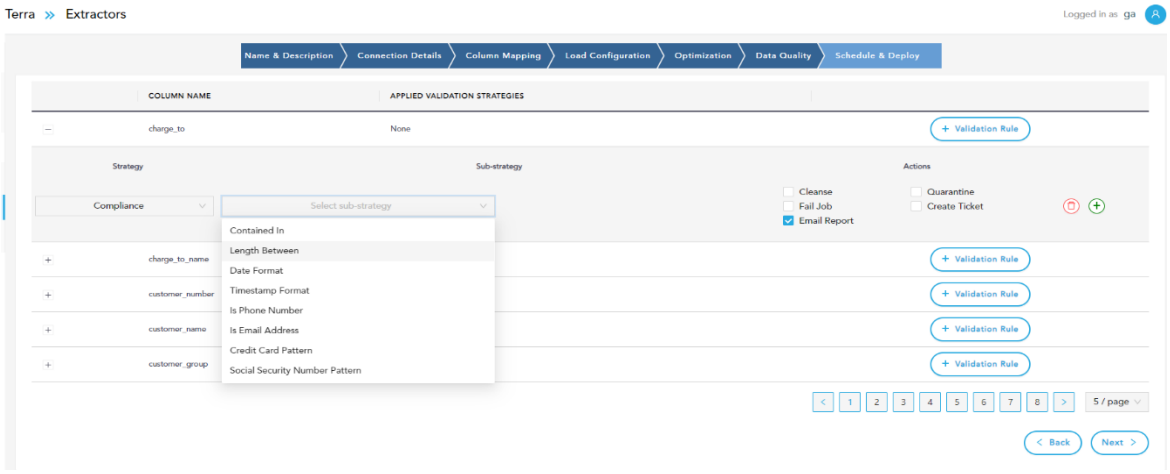


4. Enter Partition details /cluster key details and Enter compression details and file format details. User can select the column to partition the data and also user need to select compression type as we have different compression types line snappy, gzip, lz4 etc.. and file format csv, avro, ORC



5. Apply data quality checks on the columns needed. There are 4 strategy and sub strategy available for applying a validation rule
   a) Strategy - Completeness and sub strategy is not null and not empty
   b. Strategy - Consistency and not string is the sub strategy
   c. Strategy - Compliance and sub strategies are length between, contained in, date format, Timestamp format, is phone number, is email address, credit card pattern, social security number pattern
   d. Strategy- Data Profile and sub strategies are min length, max length, null values, size





6. Schedule the extractor for deploying it. User have two options to schedule, either time based or event based

   a) Time based:

      Once: Once works like the user can schedule a job only for once. That means job will run only for one time to that at which time the job was scheduled

      Recurring: If user wants tot trigger the job for some particular interval of time then user will use recurring

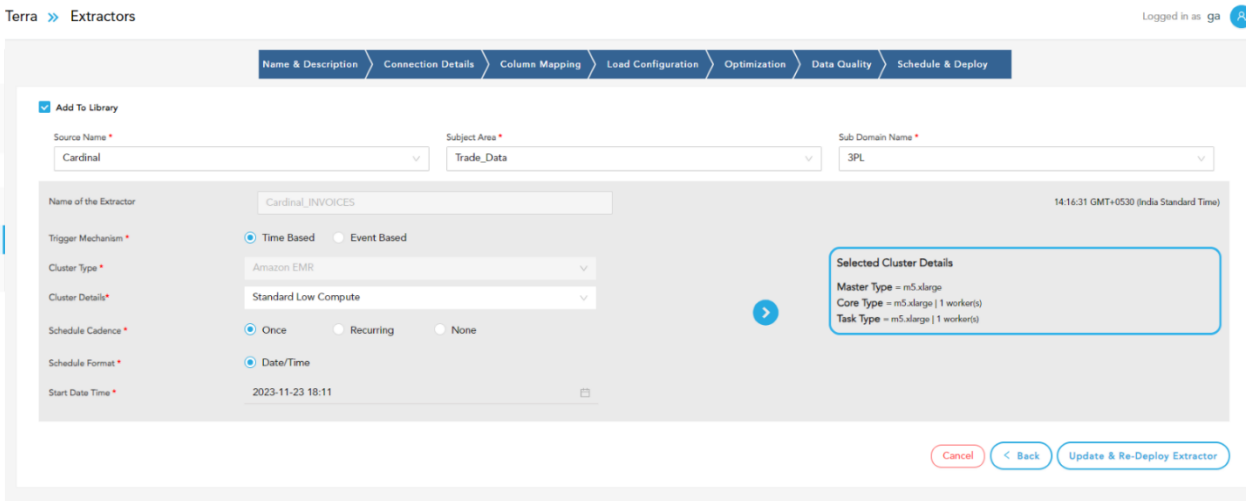
      None: None means the dag will be created but the user need to again click on RUN NOW to run the job

   b) Cluster type and details:

      User can't change the cluster type either it is emr or databricks but user can select the cluster details like low compute, medium or high compute cluster. According to the selected compute the user will be able to see the cluster details like task type, master type and core type.

   b) In schedule and deploy page user can see the extractor name which was provided at the starting of creation of extractor.

7. Once user click on deploy extractor then extractor will be deployed and the dag will be created in Airflow server.



Pre-built Extractors

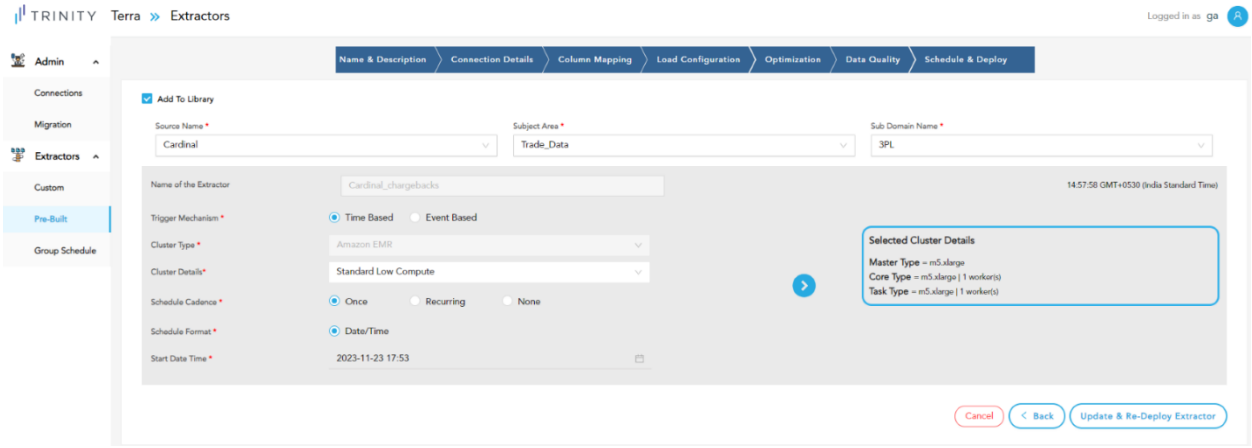
Pre-built extractors are designed to handle the extraction of data from specific sources without users having to create custom extraction processes from scratch.

In Pre-built extractors, there is an option to select the vendor, subject area, and sub-domain. Based on the inputs, users can filter the data in pre-built extractors.

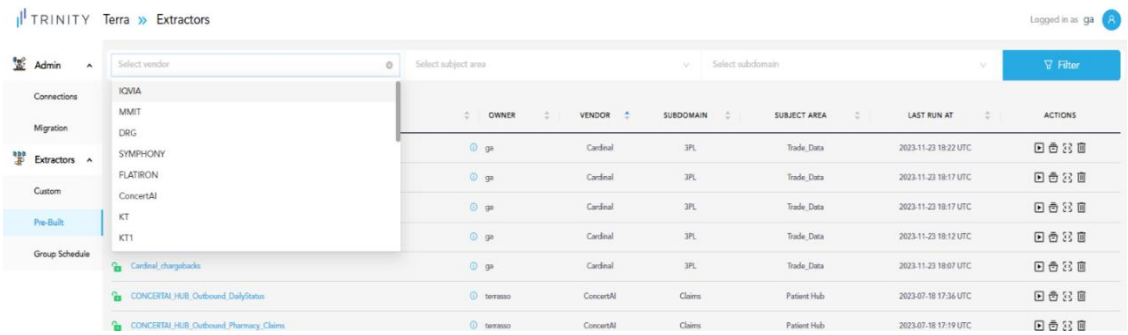
Steps for Creating a Pre-built Extractor

Select Extractors è Custom è Create a new Extractor

- Follow all the steps as per custom extractor. In the last step click on the checkbox for add to library to enter the vendor name, sub domain and subject area and schedule it



- Check for the extractor to be listed in pre-built extractors
- Searching for prebuilt extractors can be done by using the filters vendors, subject areas, sub domain



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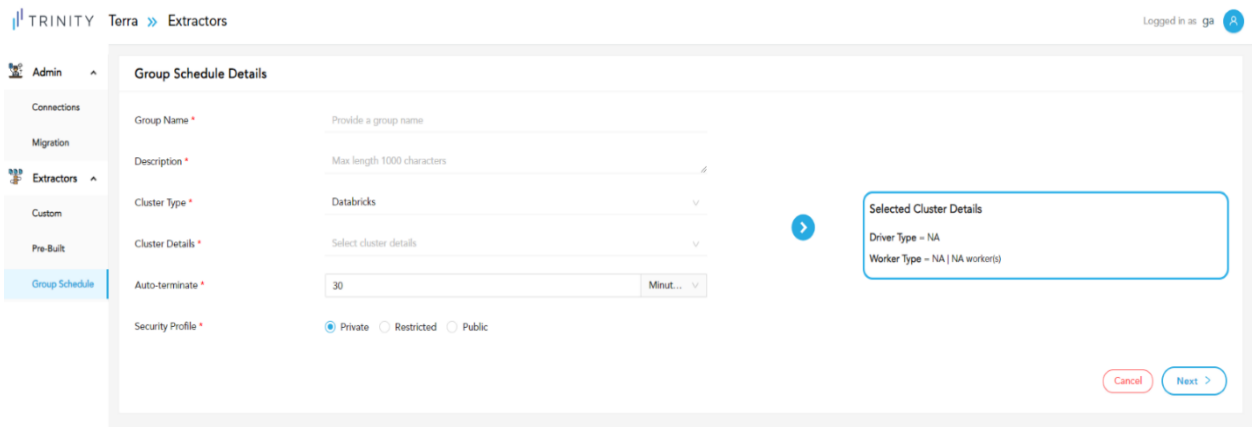
Group – Schedule

Group schedule is used to schedule extractors, pipelines, and all jobs in a single DAG.

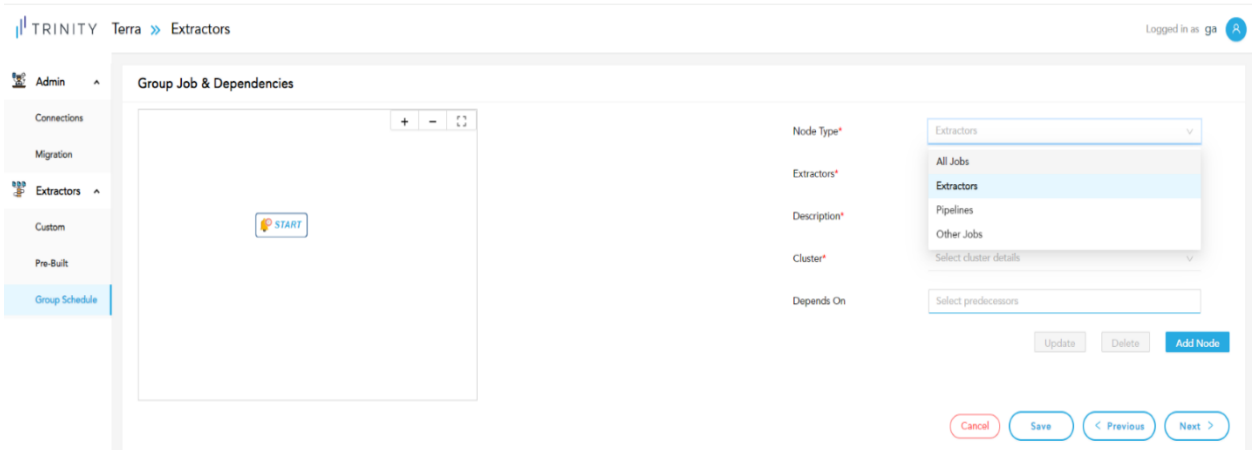
Steps for Creating a Group - Schedule

Select Extractors è Group Schedule è Create a new group

1. Enter the group name, description and choose the cluster type, cluster details, Auto terminate and security profile and click on next

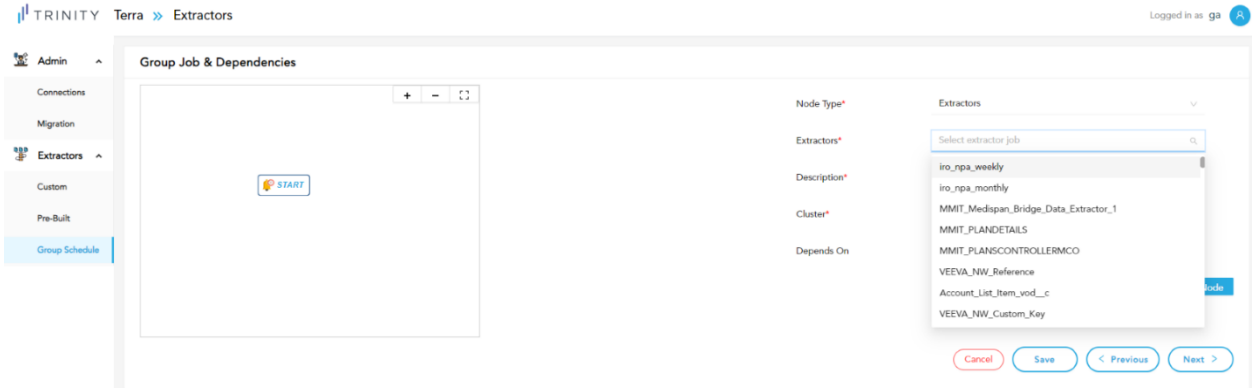


2. Select the node type among All jobs, Extractors, Pipelines and Other jobs



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3. Select the any one job from All jobs/Extractors/Pipelines/Other jobs



4. Enter the description and choose the cluster and click on add node to add the first node
5. While adding the second node choose the depends on option to schedule it parallelly or sequentially

Admin Page

In the Admin tab, the "Connections" feature establishes links with different sources, storing information in RDS (Relational Database Service). Clicking "Create New Connection" offers options for various data source systems, including RDBMS, Salesforce, Azure, AWS, Linux, Databricks, and Snowflake.

Steps for Creating Connections

Select Admin è Connections è Create a new connection

1. Enter connection type, connection name and Connection description
2. Based on the connection type chosen, user should enter the connection parameters required for connection setup
3. Connection can be added using Add connection button

Migration

The "Migration" functionality is used for importing and exporting Extractors.

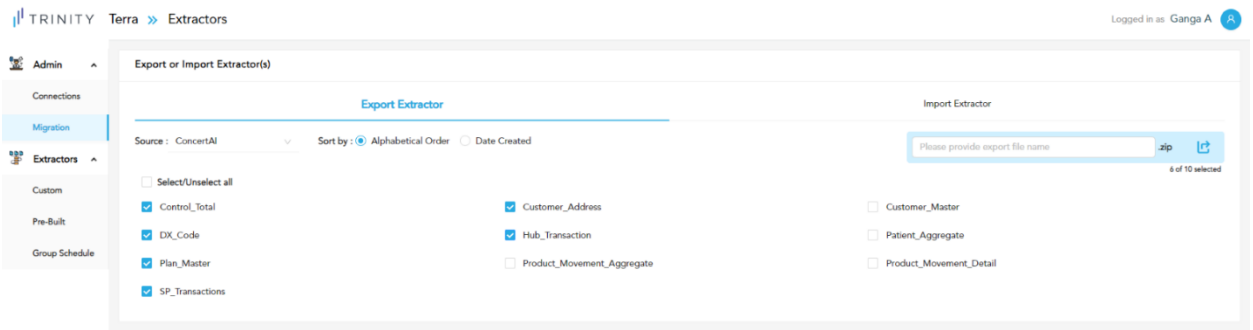
Export Extractors

Export Extractors allows users to export extractors based on their selections. The source dropdown displays a list of sources, simplifying the process for users to locate and export their desired extractors. Users can export the list of extractors in zip format by providing an export file name.

Steps for exporting extractors:

Select Admin è Migration è Export Extractors

1. Click on source to choose the list of extractors and click on the checkbox to choose the extractors to be exported
2. Enter the Export file name in the text box provided
3. Click on the download symbol to export the extractor zip file to your local memory of computer you are using



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Import Extractors

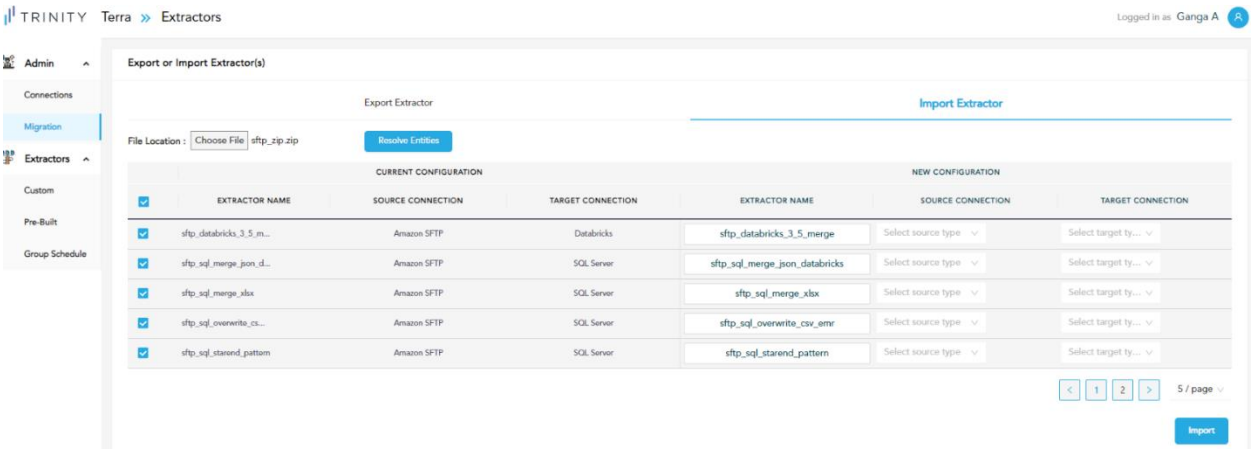
Import Extractors is used to import an extractor by choosing a zip file from the local system.

Users select a zip file from the local system and click "Resolve Entities." It displays the current configuration details of the file the user uploaded, such as the extractor name, source, and target connection. It also displays new configuration details where users can edit the extractor name. Click "Import" to complete the import process. Users can check the imported extractor on the custom page by searching with the extractor name in the search bar.

Steps for importing extractors

Select Admin➔ Migration➔Import Extractors

1. Click on choose file button to choose the extractor zip file from the local system and click on ok.
2. Click on resolve entities and check the list of extractors in the zip file .The required extractors to be imported can be selected by clicking the check boxes and the source and target systems can be chosen accordingly and click on import



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